IEEE Access
Multidisciplinary ⋮ Rapid Review ⋮ Open Access Journal

Digital Object Identifier

Recovery-Aware Physical Attribution of Load and Line Events From Sparse PMUs

JORGE LUIS MAYORGA TABORDA[1], YESSICA AFRICANO RODRIGUEZ[1], MASOUD BARATI[2], PAYMAN DEHGHANIAN[3], AND DAVID FELIPE CELEITA RODRÍGUEZ[4]

[1]Universidad de los Andes, Bogotá, Colombia
[2]Department of Electrical and Computer Engineering, University of Pittsburgh, Pittsburgh, PA, USA
[3]Department of Electrical and Computer Engineering, The George Washington University, Washington, DC, USA
[4]Faculty of Engineering, Universidad de La Sabana, Chía, Colombia

Corresponding author: Jorge Luis Mayorga Taborda (e-mail: jl.mayorga236@uniandes.edu.co).

ABSTRACT After a load is restored or a line is reclosed, the machines keep moving, and the residual response can resemble a new intervention. We formulate load/line attribution from eight PMUs in the IEEE 39-bus system as change detection and isolation with switched differential-algebraic (DAE) post-change models. A bank of linearized DAE predictors compares intervention and recovery histories that start from estimated past states, integrates load severity in closed form, and keeps a restored intervention available as the cause of the continuing response. The same operators give Willsky–Jones signatures, from which we compute, before any experiment, each source's detection and isolation information, its minimum detectable change versus noise, and the spurious evidence created by forgetting the physical state at restoration. The analysis predicts detection at the first post-onset report and reset evidence at least two orders of magnitude above the alarm threshold. In an independent simulation evaluation (18 physical parents, 108 interventions, 648 paired replays) the bank attributes 644/648 interventions before restoration with no alarm in 11.8 h of normal records; resetting the state lowers timely attribution from 108/108 to 75/108. Re-run with unchanged code, the bank tolerates a twentyfold increase of the calibrated noise; independent per-PMU noise of 0.05% TVE suppresses most load attributions although the angles still carry at least 36 times the alarm evidence, which places the limit in the noise model rather than in the physics. The single-model bank processes reports faster than the 30-report/s input.

INDEX TERMS change detection and isolation, differential-algebraic equations, event attribution, phasor measurement units, physics-informed inference, power system dynamics.

I. INTRODUCTION

AFTER a load is restored or a line is reclosed, the machines continue moving. A second intervention may begin before that response has decayed. With only eight PMUs in a 39-bus network, a change in a measured signal may therefore indicate a new source, recovery from an earlier source, or an imperfect physical prediction. The task is to distinguish these explanations using only the reports available at the decision time.

Physical restoration is not electrical recovery. A detector that returns to a flat reference after OFF can issue another alarm for the remaining oscillation, while suppressing alarms during recovery can hide a real intervention. In statistical terms, the post-change law of a second intervention depends on the history of the first. Classical change detection and isolation treats a single change of an otherwise stationary law [1]–[3], and the generalized-likelihood approach of Willsky and Jones detects jumps in a fixed linear system [4]. Power-system outage detectors compare network hypotheses under quasi-static or linearized models [5]–[7], and trajectory libraries diagnose faults from sparse PMUs [8]. None of these formulations makes a restored intervention compete, as a physical cause, with later interventions in a switched DAE network.

We address this gap with three contributions.

1) *Recovery-aware attribution as switched-DAE change detection and isolation.* Candidate histories specify intervention, source, onset, severity interval and restoration time. Their predictors are reduced from the network DAE, start from estimated past states and keep the state through restoration. Severity is integrated in closed form, and separate active and cause weights separate the present intervention from the originating cause of the continuing response (Section IV).

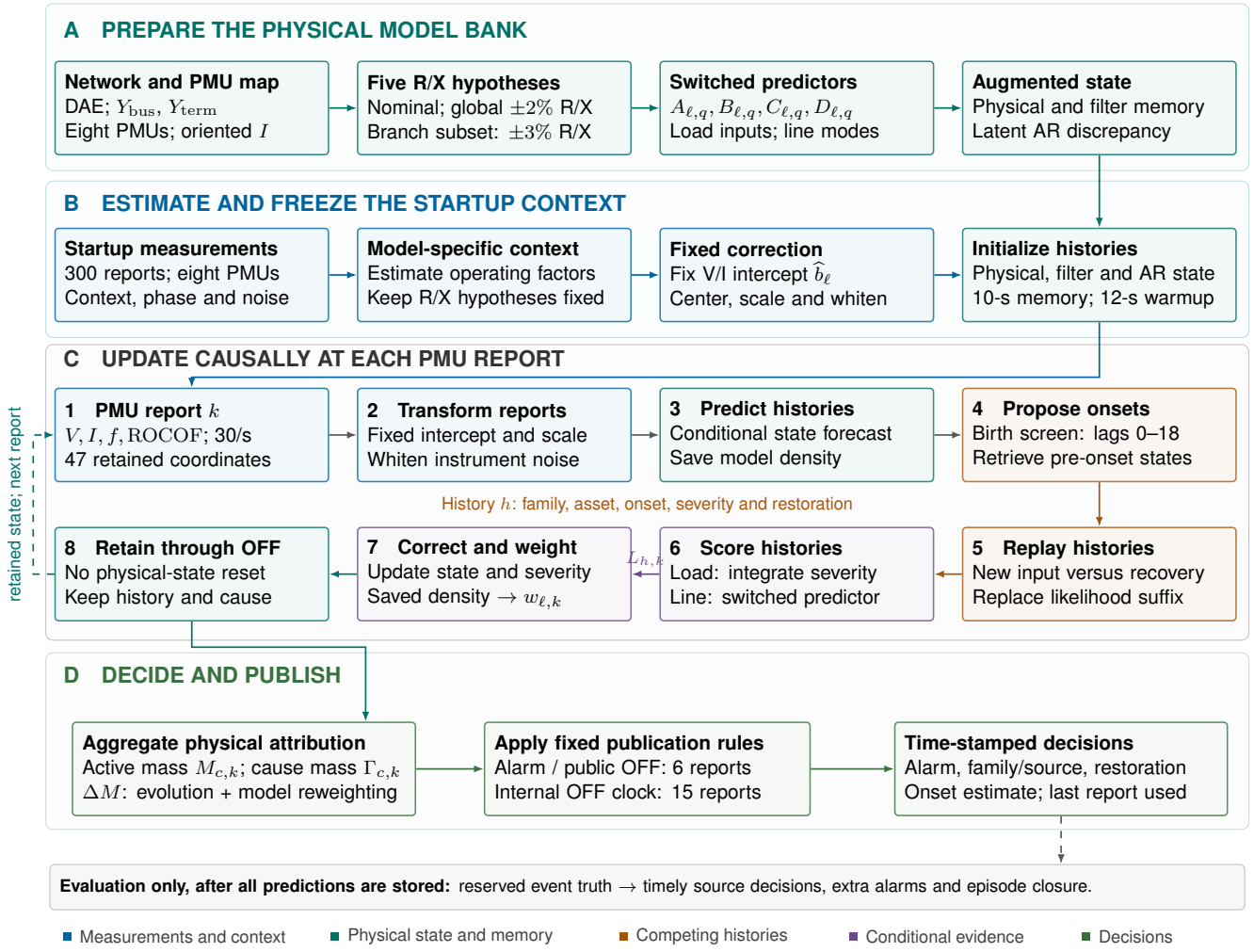


FIGURE 1. Implemented physical-history attribution: model preparation (A), startup context (B), per-report inference (C) and publication (D). The bank predicts each report before correcting any state and stores the network-model evidence before proposals are selected. A replayed candidate replaces a likelihood suffix, so each report contributes once. The feedback loop retains physical and instrumental memory through restoration. Event truth is used only after all predictions are stored.

2) *Physics-derived detectability and the cost of forgetting.* Because each history is linear-Gaussian given its mode path, its Willsky–Jones signature follows from the network operators. We compute the detection and isolation information of all 18 candidate sources, the minimum detectable load change as a function of PMU noise, and the spurious evidence created by resetting the physical state at restoration. These quantities are obtained before any campaign and are then checked against measured delays, a controlled state-reset study and a noise-stress test with unchanged code (Section V).
3) *An independent causal evaluation with explicit failure accounting.* On 18 new physical parents (108 interventions, 648 paired replays, 11.8 h of normal records) we separate detection, timely source attribution, unmatched alarms, restoration and computation time, and we report the published quickest-change detector of Chen *et al.* and two learned adapters at their original operating points (Section VII).

The bank attributes 644/648 interventions before physical restoration and raises no alarm on normal records. The analysis explains why: under the noise model calibrated on the reference data, a 10% load step or a line opening carries between about 300 and 5×10^7 times the evidence required for an alarm within one second. The errors that remain arise from network-model mismatch and from the publication rules, not from missing information. The same analysis identifies the conditions under which this ceases to hold. The supplement documents the implemented filter, the complete outcomes and the reproducibility contract.

II. RELATED WORK

A. CHANGE DETECTION AND ISOLATION IN DYNAMIC SYSTEMS

Sequential detection with a prior on the change time leads to the Shiryaev statistic, whose posterior-probability threshold

is optimal for a geometric change-time prior [9]. Detection *and isolation* among several post-change laws was formalized by Nikiforov [1] and Lai [2]; asymptotically, the delay for a given error level is inversely proportional to the Kullback–Leibler information separating the true post-change law from its competitors [3]. For linear state-space systems, Willsky and Jones introduced signatures of jumps propagated through a Kalman filter and estimated their magnitude by generalized likelihood ratios [4]. Separating a constant bias from the state, as Friedland did for recursive filtering [10], gives the affine state–severity representation used in our load histories. Interacting and multiple-hypothesis filters propagate mode-conditioned states and mode-sequence histories with pruning and merging [11], [12]. Our method inherits these mechanisms. Its post-change models, however, are switched DAE operators of a power network, and a restored intervention remains a competing cause of later observations.

B. PHYSICAL INFERENCE OF GRID EVENTS

Unknown-input and DAE estimators recover electrical states despite incomplete models or exogenous inputs [13]–[15]. Functional observability and descriptor-model diagnosability distinguish recovering a state quantity from identifying the input that produced it [16]–[19]. Phasor-angle hypotheses support outage identification [5], Bayesian outage inference with uncertain states [20], and quickest line-outage detection [6], [7]. Anguluri *et al.* recover sparse forcing locations and inputs of linear dynamics by group LASSO [21], and sensor placement determines which sources are distinguishable [22]–[24]. Warner *et al.* compare simulated voltage/current fault trajectories over a moving onset window [8] and derive error/delay guarantees for an adaptive matrix CuSum [25]. Trajectory sensitivities and Gaussian evidence have been combined for parameter inference in power-system DAEs [26], [27]. We use the same ingredients, DAE linearization, Gaussian evidence and onset search, to compute information-theoretic detectability of each catalog source and to test it against the attribution outcomes.

C. LEARNED DIAGNOSIS FROM SPARSE MEASUREMENTS

Learned and knowledge-driven representations identify PMU disturbances and assets [28]–[30]. SHRED reconstructs IEEE-39 dynamic states from a few PMUs [31], and Dey *et al.* localize distribution topology changes with hardware-in-the-loop validation [32]. These methods learn the source map from labeled trajectories; the physical bank derives it from the network equations. We report two ExtraTrees adapters trained on the same 18-source catalog, whose limitations are discussed with the results.

III. PROBLEM AND MEASUREMENT MODEL

At report k, inference uses measurements through $t_k = k/30$ to decide whether a physical intervention occurred. Detection concerns its presence; classification selects a physical operator; localization selects a bus or branch. These outputs can become reliable at different times. We record the estimated physical onset $\widehat{\tau}$ separately from the alarm and diagnosis emission times.

Figure 2 introduces the broader measurement phenomena using the reference record released with the SGSMA 2026 competition. Faults, line changes, generation changes and load changes alter the physical network; missing or corrupted reports alter observation integrity. The present causal test addresses load changes and line openings with restoration. The other panels provide context, not additional validated event classes.

The network has 39 buses, but the prescribed PMU set is

$$\mathcal{P} = \{2, 5, 6, 10, 19, 22, 29, 39\}. \tag{1}$$

Voltage is measured at each PMU bus. Current is measured at a specified branch terminal, which can differ from that bus. The causal implementation uses 47 coordinates: eight voltage magnitudes, eight voltage angles, eight current magnitudes, seven current angles, eight absolute frequencies and eight ROCOFs, at 30 reports/s. BUS39 current angle is excluded because its interpretation remains partially validated. The BUS5 current terminal retains the provisional historical 5–6 assignment. These are declared measurement assumptions, not identities inferred from classification accuracy.

The exported magnitudes are in V and A, angles in degrees, frequency in Hz and ROCOF in Hz/s. Internal centered coordinates do not replace the absolute frequency contract. These are reporting-rate phasor measurements; waveform harmonics or THD are not measured by these channels.

The causal stream is exported from one balanced complex phasor per PMU: phase A supplies the voltage and current magnitudes and angles. The CSV exporter constructs B and C by equal magnitudes and angle offsets of -120° and $+120^\circ$. Retaining A therefore loses no independent phase information in these synthetic records; it is not measured positive-sequence extraction or a model of unbalanced faults. The vector is $y_k = (|V_A|_{\mathcal{P}}, \angle V_{A\mathcal{P}}, |I_A|_{\mathcal{P}}, \angle I_{A\mathcal{P}\setminus\{39\}}, f_{\mathcal{P}}, \dot{f}_{\mathcal{P}})_k$. The supplementary measurement table gives all eight current-terminal assignments, polarities, gains and phase offsets. Products of voltage and current measured at different terminals are power proxies.

The physical model is

$$\dot{x} = f_q(x, z, u; \theta), \qquad 0 = g_q(x, z, u; \theta), \tag{2}$$

$$\begin{aligned} h_k &= H_q(x_k, z_k, u_k), \\ \eta_k &= F_M(\eta_{k-1}, h_k; \theta_M), \\ y_k &= G_M(\eta_k, h_k; \theta_M) + b + \varepsilon_k. \end{aligned} \tag{3}$$

Here x contains machine and control dynamics, z the algebraic network variables, q the topology/intervention mode, and θ_M the measurement convention. H_q selects electrical phasors; η_k retains the causal phase history used for frequency/ROCOF and the measurement-filter state. The observation thus depends on an augmented state. Bus frequency comes from phase evolution, not a machine's rotor speed. Instrumental noise is applied according to the declared generator contract.

TABLE 1. Closest inference formulations. Only the quickest-change detector of Chen *et al.* and the two ExtraTrees adapters were executed on the common evaluation data.

Route	Mechanism	Evaluated target	Role in this study
Willsky–Jones GLR; Nikiforov/Lai	Filter signatures; detection–isolation statistics	One jump or one change in a fixed system	Information analysis (Section V)
IMM / M3H	Mode-conditioned filters, history pruning and merging	Hybrid switching-state estimation	History-bank mechanism
Chen / Rovatsos	Network likelihoods, sequential tests	Line outages	Chen executed on persistent openings
Warner high-fidelity	Simulated V/I paths, onset search	Short-circuit line candidates	Closest trajectory precedent
DAE / stochastic-state estimation	Physical state and parameter models	States, inputs or parameters	Components
Learned PMU diagnosis	Features or learned representations	Event families and locations	Two ExtraTrees adapters executed
This work	Switched-DAE histories with recovery	Successive catalog load/line events	Attribution, alarms, detectability

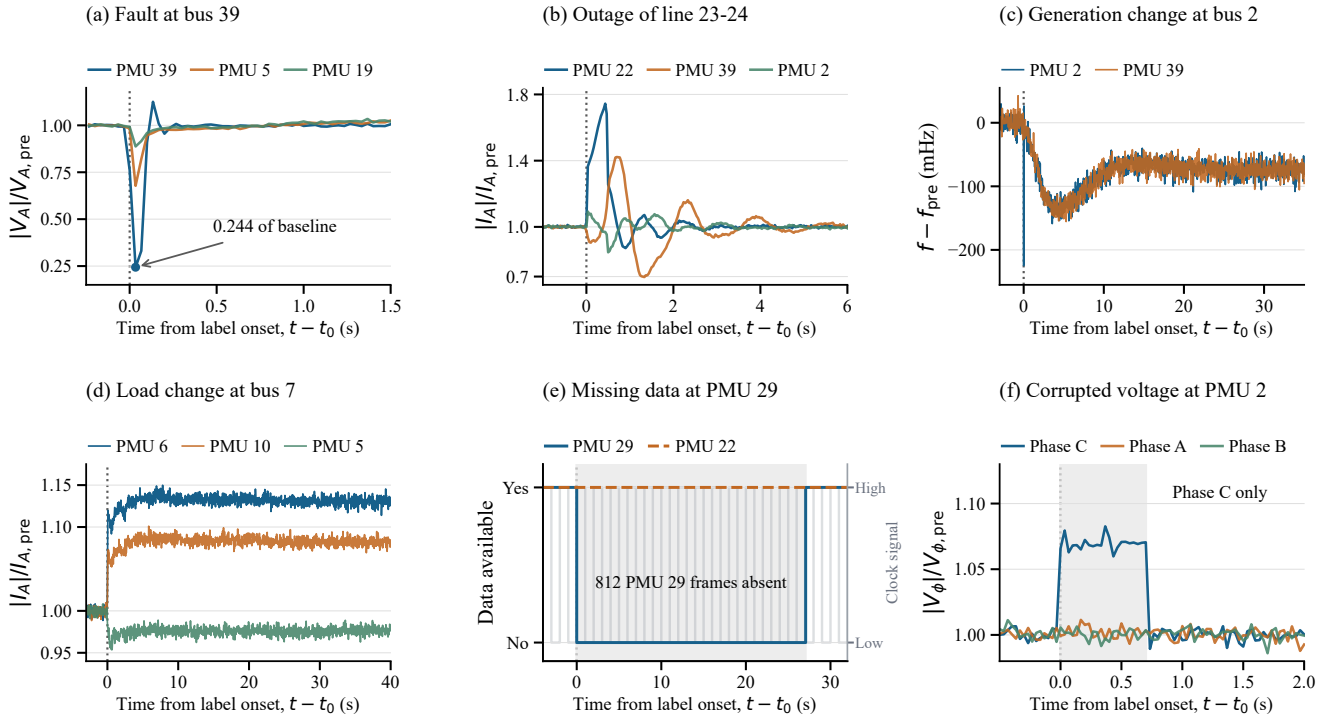


FIGURE 2. Representative signatures from the archived RAW0001 reference simulation of the SGSMA 2026 competition: fault, line outage, generation change, load change, missing PMU data and corrupted voltage. Panels are aligned at the labeled onset for illustration only; inference receives no such alignment. The gray clock in (e) is an illustrative reference, not a measured channel. Load and line mechanisms enter the causal evaluation; the other panels show the broader event taxonomy.

An intervention history is the discrete tuple $h = (e, c, \tau, \kappa, \mathcal{I})$: family, source, onset, restoration time ($\kappa = \infty$ for a sustained event) and, for a load, a severity interval $\mathcal{I}$. The severity $a \in \mathcal{I}$ is a continuous latent variable conditioned on h and integrated in the evidence; a line opening has no severity. A load changes its local P/Q setpoints by the common factor $1 + a$; a line opening switches its branch primitive. For the supported impulse-free switches,

$$x(\tau^+) = x(\tau^-), \qquad g_{q^+}(x(\tau^-), z(\tau^+), u; \theta) = 0. \quad (4)$$

The algebraic voltage/current response may jump even though dynamic states remain continuous. Physical restoration reverses the intervention operator. It does not return the machines to their pre-event state, so electrical recovery can continue after OFF.

Switch-aligned reports use the consistent post-switch algebraic solution. The local reduction requires an invertible algebraic Jacobian after fixing the angle reference; the catalog excludes islanding line alternatives. Successful solves and local residual checks validate the admitted cases, not global regularity for arbitrary switching topologies.

The generator stores event schedules and true states sepa-

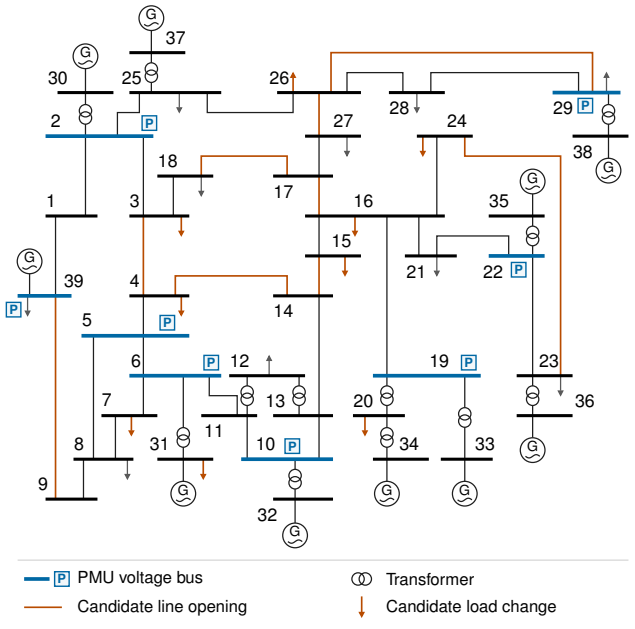


FIGURE 3. IEEE 39-bus single-line diagram: 46 branches, 12 transformers and 10 generators. Blue bars and P marks locate the eight voltage PMUs. Orange branches and load arrows identify the nine line and nine load candidates; gray arrows denote other loads. Current terminals follow the separate measurement contract. All routes are orthogonal.

TABLE 2. Scope of the executed causal component. The source is an intervention asset; it is not necessarily a measured PMU bus.

Case	Physical operator	Executed claim
Normal	No intervention	Monitor after startup
Load	Local P/Q factor; optional restoration	Nine candidate buses
Line	Branch open; optional reclose	Nine candidate branches
Fault/generation	Different shunt/input/trip operators	Outside this causal test
Missing/bad data	Measurement-integrity mechanism	Outside the stream test
Unknown	Unresolved diagnosis	Not a new physical family

rately from the PMU stream. Inference receives the nominal network, permitted calibration and reports through the current time. It receives no source, switch time, true R/X realization or solver state. Initial operating context and noise scale are estimated from 300 startup reports; monitoring begins after the inherited 12-s warmup. The startup contract assumes no intervention in that prefix.

IV. RECOVERY-AWARE PHYSICAL ATTRIBUTION

Figure 1 describes the implemented computation. The inference engine uses a conditional history bank of DAE-derived switched state-space predictors. The nonlinear simulator generates observations; inference does not solve a new full nonlinear simulation for every candidate at every report.

Physics enters the candidate dynamics, input directions and oriented measurement map. Statistical filtering represents uncertain states and discrepancy around those predictions. Thus the evaluated method is model-based inference; it does not train a neural network with a physics loss or learn a source-label classifier. A good residual fit alone is insufficient: competing histories must also yield a timely source decision.

A. CONSTRUCT THE PREDICTOR FROM THE NETWORK

Kirchhoff constraints and complex power connect sources to remote PMUs:

$$I_{\text{bus}} = Y_{\text{bus}} V, \quad I_{term,r} = Y_{term,r} V, \quad S_i = V_i I^*_{bus,i}. \tag{5}$$

Terminal rows retain orientation, line charging and transformer ratios. A measured terminal current is not a nodal injection. A line opening changes its branch primitive and terminal map; a load changes its admitted P/Q input direction.

Each network hypothesis is linearized at the operating point $(\bar{x}_\ell, \bar{u}_\ell)$ determined by its prefix-fitted load and dispatch factors. For topology q, the constructor holds $\bar{x}_\ell$ fixed and solves $g_{\ell,q}(\bar{x}_\ell, \bar{z}_{\ell,q}, \bar{u}_\ell) = 0$. On a regular index-one chart with a fixed angle reference, the derivatives at that consistent point give, suppressing ℓ,

$$\begin{aligned} \delta z &= -g_z^{-1}(g_x \delta x + g_u \delta u), \\ \delta \dot{x} &= c_q + A_q^{red} \delta x + B_q^{red} \delta u, \\ A_q^{red} &= f_x - f_z g_z^{-1} g_x, \quad B_q^{red} = f_u - f_z g_z^{-1} g_u. \end{aligned} \tag{6}$$

Here $c_q = f_q(\bar{x}, \bar{z}_q, \bar{u})$ can be nonzero after a line opening even though the differential state is continuous. For the instantaneous electrical observation map $H_q(x, z, u)$, the corresponding output operators are

$$C_q^{red} = H_x - H_z g_z^{-1} g_x, \qquad D_q^{red} = H_u - H_z g_z^{-1} g_u. \tag{7}$$

The electrical offset is $d_q^{el} = H_q(\bar{x}, \bar{z}_q, \bar{u}) - H_0(\bar{x}, \bar{z}_0, \bar{u})$. Thus both the changed algebraic solution and the changed terminal map enter the prediction. These operators propagate an unmeasured-source perturbation to the selected voltage and oriented-current channels. Network connectivity enters through the branch stamps in g_q and H_q; eliminating unmeasured algebraic variables generally produces dense reduced operators. The graph therefore constrains the prediction without a learned graph embedding. A changed terminal operator also contributes directly through H_q, not only through the changed voltage. The implementation discretizes the admitted switch modes and adds causal phase-history operators for frequency and ROCOF. Inertia, controls and load law remain those of the fixed plant.

Exponential discretization integrates c_q over each mode interval; composition handles a switch within a report. After instrumental augmentation, the resulting offsets enter $b_{\ell,q}$ and $d_{\ell,q}$ in

$$\begin{aligned} \xi_{k+1} &= A_{\ell,q}\xi_k + b_{\ell,q} + B_{\ell,q} a + w_k, \\ r_k &= C_{\ell,q}\xi_k + d_{\ell,q} + D_{\ell,q} a + v_k. \end{aligned} \tag{8}$$

Here r_k contains reference-corrected, scaled and instrument-whitened PMU coordinates, ℓ is a fixed network hypothesis, and ξ_k contains physical deviation states, phase history and predicted measurement-filter memory. Observed colored-instrument filters whiten measurements; their corresponding predictor memory is deterministic. Additional latent AR states represent persistent discrepancy, with preset quiet/load/line/recovery covariance and drift regimes. This statistical discrepancy model is distinct from the plant equations. v_k denotes the remaining innovation. A load has scalar severity within its specified P/Q direction. A line uses a switched operator rather than a fictitious load-amplitude fit. States from different equilibria are not interpreted as an absolute rotor-state mixture. The supplementary section "Implemented Predictor and Conditional Uncertainty" gives the affine construction and its code correspondence.

B. ESTIMATE CONTEXT FROM THE STARTUP PREFIX

Each network hypothesis uses the same 300 startup reports to estimate operating factors, reference phase and noise scale. Voltage/current coordinates receive a constant discrepancy correction,

$$\widehat{b}_\ell = \bar{y}_{0:299} - y_\ell^{ref}, \qquad b_{\ell,k} = \widehat{b}_\ell \quad (k \geq 300). \tag{9}$$

Angular differences respect wrapping and the common reference. Frequency and ROCOF use empirical prefix centers. The correction remains fixed so it cannot track a later load step and subtract its signature. This estimates an observation discrepancy, not the true impedances. Prefix estimates enter by substitution rather than through a full joint parameter posterior.

C. COMPARE UNKNOWN-ONSET HISTORIES

The method retains ten seconds of past conditional states. Its birth screen searches lags of 0–18 reports and proposes selected onsets; it does not enumerate every onset in the 300-report history. Each history specifies family, asset, onset, restoration mode and severity. It starts from the stored conditional state before its proposed onset. That state already contains any ongoing recovery; an alarm does not reset the predictor to equilibrium.

For a retained load history, conditional Gaussian filtering yields a residual e, severity response s and covariance S. Unknown severity has a truncated-Gaussian distribution inherited from previous reports:

$$L_{h,k} = \int_l^u \mathcal{N}(e; sa, S)\, \mathrm{TN}_{[l,u]}(a; \mu, v)\, da. \tag{10}$$

With Φ denoting the standard normal CDF, define

$$\begin{aligned} v' &= (v^{-1} + s^\top S^{-1} s)^{-1}, \\ \mu' &= v'(\mu/v + s^\top S^{-1} e), \\ Z(\mu, v) &= \Phi((u-\mu)/\sqrt{v}) - \Phi((l-\mu)/\sqrt{v}). \end{aligned} \tag{11}$$

The implemented local integral follows by completing the square:

$$\begin{aligned} \log L_{h,k} = \log \mathcal{N}(e; 0, S) + \tfrac{1}{2}\log(v'/v) \\ + \tfrac{1}{2}(\mu'^2/v' - \mu^2/v) + \log \frac{Z(\mu', v')}{Z(\mu, v)}. \end{aligned} \tag{12}$$

These standard identities are exact for the stated local conditional model. Adaptive proposals, pruning and covariance approximations make the complete bank approximate.

As in separate-bias filtering [10], the conditional state mean is represented as $m_h + d_h a$; d_h is the Willsky–Jones signature of a unit severity propagated through the filter [4]. To make the filtering approximation explicit, let $T_o, J_o, g_o, u_o, j_o, v_o$ denote the one-report transition and observation operators for switch o. The code forms $e = r_k - J_o m_h - j_o$ and $s = J_o d_h + v_o$. With the shared covariance P_ρ for the current discrepancy regime ρ,

$$\begin{aligned} S_o &= R + J_o P_\rho J_o^\top, & K_o &= T_o P_\rho J_o^\top S_o^{-1}, \\ m_h^+ &= T_o m_h + g_o + K_o e, & d_h^+ &= T_o d_h + u_o - K_o s. \end{aligned} \tag{13}$$

The amplitude distribution is updated by the same integral. Its truncated variance $V_{a,h}$ differs from the Gaussian parameter v_h. The affine conditional representation retains state–severity dependence:

$$\begin{aligned} \mathrm{Cov}(\xi, a \mid h) &= d_h V_{a,h}, \\ \mathrm{Var}(\xi \mid h) &= P_\rho + d_h V_{a,h} d_h^\top. \end{aligned} \tag{14}$$

Conditioning on past reports is implicit. The shared P_ρ solves the nominal-operator Riccati equation of regime ρ; S_o and K_o are then formed for each switched operator o. This is a steady-state approximation, not an exact covariance recursion along each switched history. At closure, the filter retains the augmented-state mixture mean and records the amplitude/mean spread that it discards. The supplement separates these approximations from the exact local amplitude integral and states the covariance needed for a full mixture.

The rolling screen proposes histories; it is not another independent observation. Replaying a suffix replaces its original no-new-event suffix likelihood. The sequential update therefore counts each report once, rather than repeatedly accumulating overlapping ten-second windows.

D. REPRESENT FIXED NETWORK UNCERTAINTY

The bank contains nominal, global $\pm 2\%$, and selected-branch $\pm 3\%$ paired R/X hypotheses. Parameters stay constant within a hypothesis. Its outer weight is updated with the one-step predictive density

$$w_{\ell,k} = \frac{w_{\ell,k-1} p_\ell(y_k \mid \mathcal{F}_{k-1}, \widehat{b}_\ell)}{\sum_j w_{j,k-1} p_j(y_k \mid \mathcal{F}_{k-1}, \widehat{b}_j)}, \tag{15}$$

$$p_\ell(y_k \mid \mathcal{F}_{k-1}) = \sum_{h'} \pi_{\ell,h',k-1} \sum_{h \in \mathcal{T}(h')} p_\ell(h \mid h')\, L_{\ell,h,k}. \tag{16}$$

The transition set $\mathcal{T}(h')$ contains continuation, restoration at either half-report phase for an eligible active history, and a

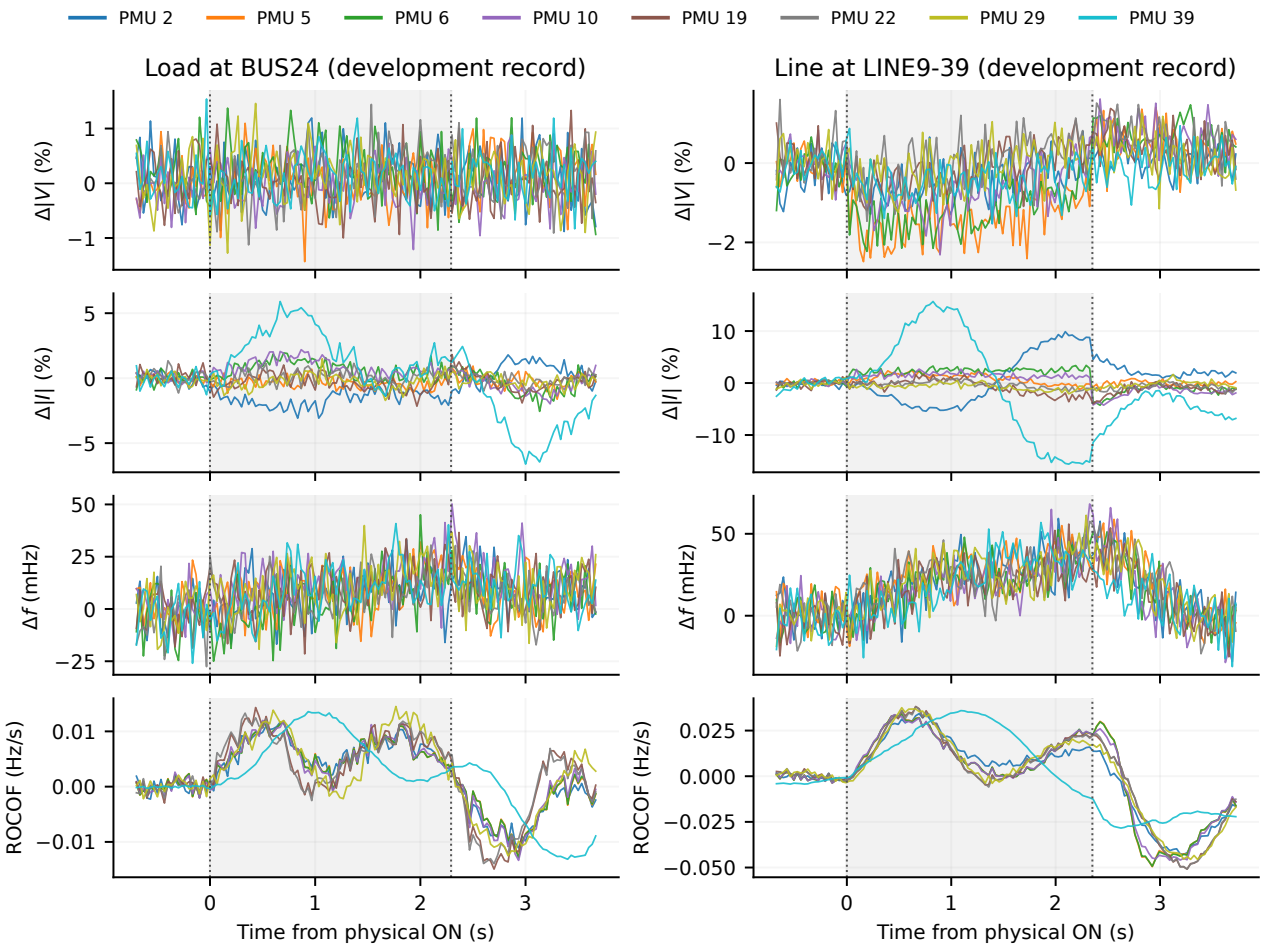


FIGURE 4. Simulated observations supplied to inference. One fixed development parent provides a load and a line example at all eight PMUs. Shading marks physical ON–OFF for evaluation only. Magnitude changes use the local pre-event center; frequency is centered for display while its exported channel remains absolute Hz. Source, severity and operating context affect the curves, so they are not universal class templates.

birth of every catalog cell and severity interval at either phase from the no-intervention history; the transition probabilities sum to one, which the implementation asserts at every report. The density is evaluated before any state is corrected and before retrospective proposals are selected. Onsets proposed later for earlier reports change the inner history posterior through suffix replay but do not re-weight w_ℓ; this is the one point where the outer and inner recursions differ from a single joint posterior. Startup-fitting reports are not scored again. The mixture covers a finite set of network alternatives rather than a continuous impedance posterior.

E. KEEP RECOVERY AS A COMPETING PHYSICAL EXPLANATION

Restoring an input or topology at κ does not restore the pre-event state: $x(\kappa^+) = x(\kappa^-)$. On a regular local post-restoration chart, the difference from a same-context no-new-intervention trajectory obeys

$$\begin{aligned} \delta x_h(t) &\simeq \Phi_0(t,\kappa)\delta x_h(\kappa), \\ \delta y_h^{\mathrm{net}}(t) &\simeq C_0(t)\Phi_0(t,\kappa)\delta x_h(\kappa), \quad t > \kappa. \end{aligned} \tag{17}$$

Here y^{net} denotes voltage/current coordinates before the instrument map. Instrumental memory is retained as well. This local interpretation of known dynamics explains why the reference after OFF is a predicted trajectory.

For conditional history mass $\pi_{\ell,h,k}$, define active and originating-cause masses for an admissible family/source cell c:

$$\begin{aligned} M_{c,k} &= \sum_{\ell} w_{\ell,k} \sum_{h:c(h)=c} \pi_{\ell,h,k}\, \mathbf{1}\{h \text{ active at } k\}, \\ \Gamma_{c,k} &= \sum_{\ell} w_{\ell,k} \sum_{h:c(h)=c} \pi_{\ell,h,k}. \end{aligned} \tag{18}$$

A retained history can stop being active while its cause still explains recovery. Later interventions compete with those histories. The implementation stores 18 admissible cells, not every family at every bus. After confirmed episode closure, explicit histories are compressed into the retained state mean. Cause masses describe the current bank, not a permanent ledger of all past interventions.

Writing the conditional active mass as $M_{\ell,c,k}$ yields

$$\Delta M_{c,k} = \sum_{\ell} w_{\ell,k}(M_{\ell,c,k} - M_{\ell,c,k-1}) + \sum_{\ell}(w_{\ell,k} - w_{\ell,k-1})M_{\ell,c,k-1}. \quad (19)$$

The first term is within-model evolution and the second network-model reweighting. Because of pruning and the steady-state covariances, M and Γ are approximate attribution weights, not calibrated probabilities.

F. EMIT SEPARATE DETECTION, DIAGNOSIS AND RESTORATION MESSAGES

An alarm is published when the active mass $\sum_c M_{c,k}$ exceeds 0.99 for six consecutive reports; with a geometric change-time prior this is a Shiryaev-type posterior threshold [9], and the six-report confirmation suppresses brief posterior excursions. OFF is published when the restoration mass exceeds 0.99 for 15 reports. The diagnosis reports the family with the largest cause mass and its highest-mass source; episode-replacement rules handle a different source that begins before the previous episode closes. Every message records its emission time and the last report used, so a diagnosis can be correct but too late to precede physical OFF.

The nominal comparator is the same procedure with only the nominal network hypothesis.

The per-report birth and restoration hazards are $1 - \exp[-(1/60)/30]$ and $1 - \exp[-(1/30)/30]$, fixed during method development before the independent schedules. They are deliberately conservative: test interventions last 2–5 s and are separated by 3–8 s, much shorter than the prior means of 60 and 30 s. A low birth hazard raises the evidence needed for an alarm by $-\log \pi_c$; Section V shows that this margin is small compared with the available information. Birth mass is split equally between load and line families, then uniformly among sources within each family. Load severity has a zero-mean Gaussian prior with standard deviation 0.15, truncated to $[-0.2, 0.2]$ and integrated over four intervals with boundaries $-0.2, -0.1, 0, 0.1, 0.2$. Each interval is a separate history with its own weight and truncated amplitude distribution; its birth weight is the prior probability of that interval. The physical operators are shared across these four histories. For a fixed path with all four intervals retained, their weighted evidence equals the single-interval integral. This partition is a bank representation, not numerical quadrature or a piecewise physical approximation. After three reports, eligible active histories can produce OFF children at either half-report position while inheriting state and severity. After likelihood normalization, the beam retains the 96 highest-weight recent histories, 64 older histories and at least one per occupied source; their union is renormalized and discarded mass is recorded. The supplement maps each operation to its implementation and lists the complete publication configuration.

G. RELATION TO MULTIPLE-HYPOTHESIS FILTERING

For a retained history h, let $p_\ell(h \mid h')$ contain the admitted birth, continuation and restoration transitions. Before pruning, its normalized evidence update has the standard form

$$\pi_{\ell,h,k} \propto L_{\ell,h,k} \sum_{h'} p_\ell(h \mid h')\pi_{\ell,h',k-1}. \quad (20)$$

This is the mode-sequence update of IMM and M3H filtering [11], [12]. Three features are specific to the present problem: the modes are oriented electrical intervention operators derived from the network DAE, histories carry a restoration time so that a restored intervention remains a competing cause, and the output separates active intervention from originating cause through (18). Retrospective births reuse a stored pre-onset state, and confirmed closure compresses histories into the retained augmented-state mean.

H. PER-REPORT ALGORITHM AND COMPUTATIONAL BOUNDARY

Algorithm 1: causal physical-history attribution. The inputs are the nominal model, the fixed method configuration and the successive PMU reports; event truth never enters these steps.

1) Estimate operating context, common reference and noise scale from the startup prefix. Initialize conditional states and retain the eligible past states needed by retrospective proposals.
2) At report k, apply the fixed centering, scaling and instrumental transformation. Propagate each retained physical/recovery history and its conditional uncertainty to predict this report. Compute and save the outer model predictive density before current-report screening and state correction.
3) Apply the short-lag birth screen using retained past states. A proposed history starts before its selected onset; replay replaces the corresponding likelihood suffix rather than adding it twice.
4) Score candidate histories and replayed suffixes before correcting their states with the reports being scored. Integrate the allowed load severity and apply the switched operator for a line hypothesis.
5) Correct the conditional states and severity distributions; update network weights and the retained history bank using the fixed rules. Preserve dynamic state and originating cause through restoration.
6) Aggregate active mass M, cause mass Γ and ΔM. Apply the fixed alarm, diagnosis and OFF-publication rules. Store emission time, estimated onset, family/source and the last report used.

The nonlinear simulator constructs observations; this algorithm propagates local switched predictors. Decisions use only reports already received. Computation time is reported separately in Section VII.

V. PHYSICS-DERIVED DETECTABILITY AND THE COST OF FORGETTING

Each retained history is linear-Gaussian given its mode path. The information that the eight PMUs carry about every candidate intervention can therefore be computed from the predictor operators before any attribution experiment is run. This section derives three such quantities and states what they predict; Sections VII and VII-F test the predictions.

A. SIGNATURES AND INFORMATION

Let intervention i start at report 0 with severity a and let the plant follow the local switched model (8) without noise. Run the conditional filter of candidate history j, started at the same onset, on these reports. Its innovation $e_j(k)$ and severity sensitivity $s_j(k)$ follow the recursion (13). The filter of the true history at the true severity has zero innovation, so the expected log-likelihood ratio of i against j after k reports is the whitened signature energy

$$\Lambda_{ij}(k) = \tfrac{1}{2} \min_{b\in[l,u]} \sum_{r=0}^{k} \left\| e_j(r) - s_j(r)\, b \right\|^2_{S_j^{-1}}, \qquad (21)$$

where the minimization over the competitor's severity is omitted when j is a line. With j the no-intervention history, Λ_{i0} is the *detection information*; $\min_{j\neq i} \Lambda_{ij}$ is the *isolation information*. Both follow from the reduced input directions, the topology offsets c_q and d_q^{el}, the terminal map and the instrument model; no data are fitted. For a load, $\Lambda_{i0} = \frac{1}{2} a^2 G_i(k)$, with G_i the accumulated information per unit severity squared. In change detection–isolation theory the asymptotic delay at a fixed error level is inversely proportional to this information rate [2], [3].

With birth prior π_c per report and cell, the active mass of the true history at its own onset reaches 0.99 once

$$\Lambda_{i0}(k) \;\geq\; \lambda^\star = \log 99 - \log \pi_c, \qquad (22)$$

which is $\lambda^\star \approx 15$ nats for the configuration of Section IV. The minimum load change detectable within a horizon T under a noise standard-deviation multiplier α is then

$$a_{\min}(\alpha, T) = \alpha \sqrt{2\lambda^\star / G_i(T)}. \qquad (23)$$

B. THE COST OF FORGETTING THE PHYSICAL STATE

Suppose the conditional mean tracks the augmented state until an inferred closure at report k_c. If the state is carried, the innovations after k_c keep zero mean. If the physical coordinates are set to zero while the instrument and discrepancy states are kept, the no-intervention filter receives the deterministic bias

$$\bar{e}_0(k) = C_0 \Phi_0(k, k_c)\, \delta x(k_c) - \sum_{r=k_c}^{k-1} C_0 \Phi_0(k, r+1) K_0\, \bar{e}_0(r), \qquad (24)$$

where $\delta x(k_c)$ is the zeroed physical part of the augmented state, $\Phi_0(k,r) = T_0^{k-r}$ and $C_0 = J_0$ are the no-intervention operators of (13), and K_0 is its gain. This is the free recovery response (17) seen through a filter that tries to absorb it. Any birth hypothesis j that explains part of this bias gains $\Gamma_j^{sp}(k) = \Lambda_0^{reset}(k) - \Lambda_j^{reset}(k)$ over the no-intervention history, and a spurious alarm becomes possible when $\max_{j,k} \Gamma_j^{sp}(k) \geq \lambda^\star$. With carry, $\Gamma_j^{sp} \equiv 0$. The bias, and therefore the cost of forgetting, grows with the energy of the slow electromechanical modes excited by the intervention and with how strongly the PMUs observe them.

C. PREDICTIONS

We evaluated (21)–(24) with the nominal operators built by the evaluated implementation in its pre-alarm (quiet) covariance regime. A consistency check confirms that the true-history filter leaves a residual below 10^{-8} nats. Fig. 5 gives the results.

Prediction 1: detection is not information-limited under the calibrated noise. A 10% load step carries between 4603 and 6.9×10^6 nats in one second, and a line opening at least 4.9×10^6 nats, against $\lambda^\star \approx 15$. The nearest competitor is separated by a comparable margin, so detection and isolation should both occur at the first post-onset report and the alarm should be limited by the six-report confirmation, i.e. at about 0.20 s. Doubling the noise should not change the delay.

Prediction 2: a single detectability limit. Under a uniform noise multiplier, every load except bus 31 remains detectable at the smallest tested change (3%) up to a multiplier of 20; the worst of them needs 1.9% there. The bus-31 load is only 9.2 MW, and its 3% change becomes undetectable within one second beyond a multiplier of about 7.4.

Prediction 3: forgetting is costly everywhere. After every one of the 108 interventions of the controlled study, resetting the physical state gives some birth hypothesis at least 2.2×10^3 nats of spurious evidence, more than 10^2 times $\lambda^\star$; with carry the spurious evidence is zero.

These predictions are deliberately coarse. They use a single linearization, steady-state covariances and the true onset, and they ignore the proposal screen and the publication rules. Their value is that they are computed from the network equations alone and can fail.

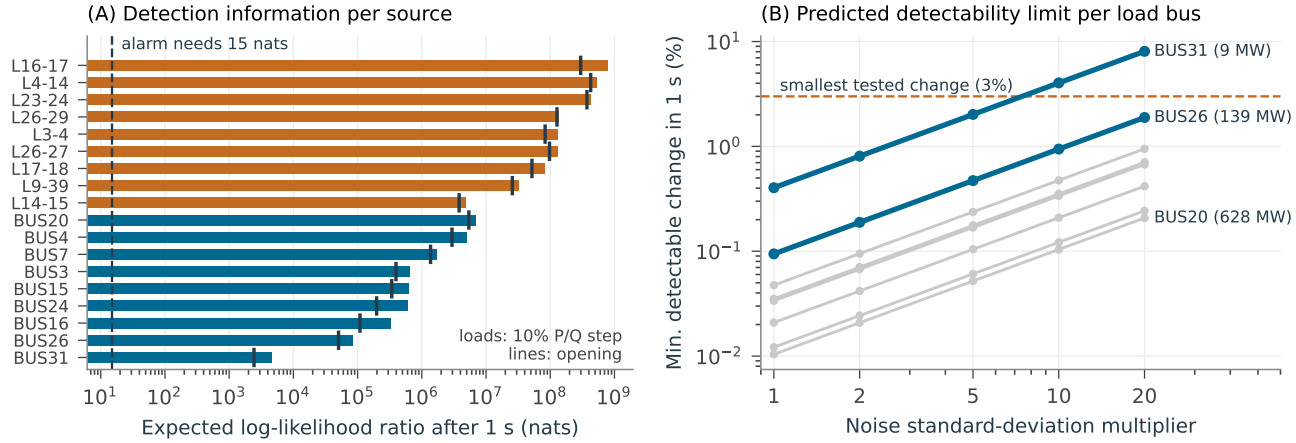


FIGURE 5. **Physics-derived detectability from the nominal switched-DAE operators, computed before the attribution campaign. (A) Expected log-likelihood ratio after 1 s against the no-intervention history (bars) and against the nearest competing source (vertical tick), for a 10% load step or a line opening; the dashed line marks the evidence $\lambda^\star$ required for an alarm. (B) Minimum load change detectable within 1 s, (23), versus a uniform multiplier of the calibrated noise; gray lines are the other seven load buses.**

VI. EXPERIMENTAL DESIGN

The primary experiment asks whether the causal bank can attribute successive load/line interventions without confusing recovery with a new event. Development, independent evaluation and an auxiliary source-resolution experiment answer different questions and retain separate denominators.

The plant is an IEEE-39 model in Julia/PowerDynamics with Sauer–Pai machines, IEEE Type I exciters, TGOV1 governors and constant-impedance ZIP loads. Each physical parent is integrated continuously, preserving differential states through ON/OFF. The PMU reporting grid is separate from solver steps: a 120-s record contains 3601 reports per PMU and a 600-s normal record contains 18001. At a switch-aligned report the export uses the post-switch algebraic value. The bus-31 candidate is the existing 9.2-MW/4.6-Mvar load at the generator bus that serves as slack only in the initial power flow; the dynamic model has no slack bus, so its load change is absorbed by machine dynamics like any other.

Instrument noise follows a stochastic model fitted to normal segments of the competition reference record: per-family vector autoregressions with a strong common factor across PMUs. Voltage-magnitude noise is about 0.4% of nominal, but the noise of angle *differences* between PMUs is only 10^{-4} to 10^{-2} degrees, because most angle noise is common to all units. This small differential noise is what makes megawatt-scale changes observable at remote PMUs (Section V); it is a property of the reference data, smaller than the idiosyncratic error expected from field PMUs [33]. Section VII-F therefore tests larger and independent per-PMU noise.

The two development campaigns each contain three event parents with six successive load/line interventions and one normal parent, all 80 s long. Schedules are paired across nominal, in-bank R/X and mixed out-of-bank R/X conditions. Each campaign therefore has 18 unique interventions and 54 event realizations. Both campaigns are now development evidence. No interval or generalization claim treats those 54 realizations as independent events.

TABLE 3. **Candidate assets fixed during method development, before the independent schedules. This is a known catalog rather than a representative sample of all network assets.**

Family	Candidate physical sources
Load buses	3, 4, 7, 15, 16, 20, 24, 26, 31
Lines	3–4, 4–14, 9–39, 14–15, 16–17, 17–18, 23–24, 26–27, 26–29

The five-model bank and its nominal version were fixed before the independent schedules were drawn. Both Extra-Trees models were trained specifically for the causal load/line catalog, using 32 earlier physical parents for training and eight for development validation. Both training sets include normal operation and all 18 candidate sources. The reference uses 846 causal features from 1-, 5- and 10-s windows; the short-window variant adds 611 features from 3-, 8- and 15-report windows. Each model has 400 trees with the same predeclared hyperparameters. They inherit feature families from the SGSMA project, but their fitted weights are distinct from the original hackathon submission. These models and their adapters were fixed before the present independent schedules; no retraining uses the current evaluation records. All methods receive the same 30-fps observations. The supplementary reproduction table records the training lineage and model hashes.

The learned adapter publishes a change after six consecutive identical argmax labels; its tree votes are uncalibrated scores. In its own development validation the reference model already attributed no load before OFF and the short-window variant four, while lines were attributed well; both were nevertheless retained unchanged under a predeclared rule. They are therefore reported as fixed adapters at their original operating points, not as the best attainable learned baseline. A retrained learner with a threshold calibrated to a common alarm budget

TABLE 4. Experimental populations and their purposes. Conditions and noise realizations repeat a physical parent rather than creating independent trajectories.

Population	Configuration	Main outcome	Interpretation
Continuous development	Three event parents, six interventions each; one normal parent; three network conditions	Timely family/source, extras and restoration	Open development evidence, not independent confirmation
Independent continuous evaluation	18 event parents, six interventions each; 12 normal parents; three network conditions and two noise scales	Paired attribution, false alarms, delays and failures	Prespecified comparison of unchanged methods
Persistent-line evaluation	Nine lines, three parent repetitions; same network/noise strata	First stopping decision, source and censoring	Eligible comparison with published line-outage QCD
Supplementary signature test	57 dictionary and 85 test trajectories; four loads; three common noise seeds	Nominal, reference correction and mixed-signature correction	Known ON/OFF and family; separate from causal accuracy
Supplementary recovery geometry	38 trajectories; four recovery histories and 12 correlated source-pair comparisons	Separation with known history and one uncertain severity direction	Supplied prior trajectory, onset, family and pair; separate from stream accuracy

is the subject of a separate study.

The published physical comparator is Chen *et al.*'s persistent single-line quickest-change detector [6]. Its reduced-network covariance is marginalized to the observed PMU subset after inversion. Parallel CUSUM statistics stop at the first threshold crossing and select a line. Our declared research-plant adapter uses nonoverlapping angle increments at 15 Hz, a startup-only scale/drift estimate and the fixed nominal line catalog. Its white-injection/DC assumptions do not describe all simulated instrumental effects. We preserve both matched-law implementation checks and model-mismatch failures; no ARL guarantee is transferred to colored noise.

The eligible common test for Chen is one persistent line opening or a normal record. Its output has no load family or restoration operator. The independent campaign therefore separates continuous load/line episodes from persistent-line first-stop decisions. Early stops, wrong sources, omissions and horizon censoring remain visible.

For continuous events, an alarm is associated temporally before predicted family/source is inspected. The first unmatched alarm after ON is eligible up to the next ON, with the inherited 30-s cap. A timely joint decision requires the correct family and asset emitted before true OFF. Unmatched alarms are extras; false restorations and open episodes are reported separately. These definitions prevent an eventual correct answer from hiding a late diagnosis.

The independent campaign specifies 18 continuous parents, 27 persistent-line parents and 12 normal parents, with three R/X conditions and instrumental-noise scales of one and two. The 171 intended physical trajectories yield 342 observed records. Conditions represent nominal parameters, perturbations represented in the bank and mixed perturbations outside it. Solver and inference failures remain in the intended denominators. Comparisons use 10,000 paired parent-cluster resamples; repeated conditions/noise do not count as independent parents. Predictions are fixed before the evaluator opens reserved truth.

TABLE 5. Held-out continuous-stream decisions. Each family has 324 replays of 54 distinct interventions; extras are unmatched alarms on event records. The ExtraTrees adapters keep their original operating points.

Method	Load	Line	Extras
Physical bank	322/324	322/324	72
Nominal bank	322/324	321/324	92
ExtraTrees	15/324	315/324	29
ExtraTrees short	30/324	322/324	16

Logical decision delay is measured on report timestamps. Wall-clock cost is recorded separately with hardware, process concurrency and one-thread BLAS settings; parallel execution of independent records does not make a within-record inference method real-time. The independent results are followed by a development example and the controlled component study.

VII. RESULTS AND CONTROLLED COMPARISONS

A. INDEPENDENT EVALUATION: TIMELY ATTRIBUTION AND ALARM COST

The five-model bank makes 644/648 timely family-and-source decisions with 72 unmatched alarms on event records, and its nominal counterpart 643/648 with 92. These are paired replays of 108 interventions in 18 physical parents, not 648 independent events. On the 72 normal records (42,336 s, 11.8 h of monitored time per method) neither bank raises an alarm, which bounds the normal false-alarm rate below 0.26 per hour at 95% confidence (Poisson). Every matched diagnosis emitted before OFF names the correct family and source; the timely misses are late or absent diagnoses. Table 5 and Fig. 6 give the family split.

The extra alarms concentrate where the network model is wrong: 68 of the 72 occur when the true R/X perturbation lies outside the five-model bank, 68 alarms in 1.08 h of out-of-bank monitored event-record time (about 63 per hour), against four in the other four strata together. Out-of-bank

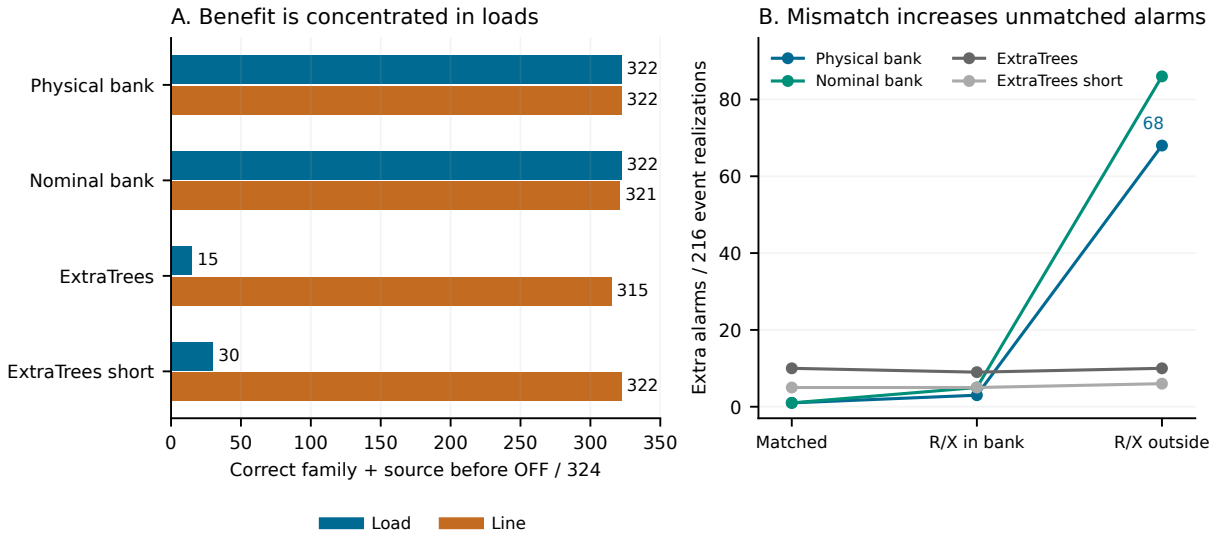


FIGURE 6. **Held-out continuous evaluation. (A) Timely correct family and source by event family, with 324 paired replays per family. (B) Unmatched alarms on event records for the two noise levels of each R/X condition (216 event replays per condition). Each method keeps its own publication rule.**

mismatch also produces the only four timely misses. Fig. 8 gives the paired parent-cluster intervals; the physical bank's median alarm delay is about 0.27 s in the matched and in-bank conditions, comparable to the learned adapters (0.19–0.30 s).

These delays test Prediction 1. For the nominal bank on matched networks, line alarms arrive after a median 0.18 s, at the six-report confirmation floor of 0.20 s less the sub-report onset phase, and doubling the noise leaves them unchanged. Load alarms arrive after 0.35 s, again unchanged by doubled noise but about 4 reports later than the available information requires. Line detection is therefore information-saturated, as predicted, while the load delay contains an algorithmic component that the information analysis exposes.

The two ExtraTrees adapters recover lines (315 and 322 of 324) but few loads (15 and 30). Fig. 7 shows that this is a coverage failure: every load diagnosis the adapters do emit before OFF is correct, but most load interventions receive none. A per-report audit on four further matched records (12 load and 12 line interventions, 1,245 and 1,143 in-event reports) locates the failure before any publication rule. During every load intervention the reference model's per-report argmax is the normal class in all reports, and the median of the largest true-class score is 0.09; during line interventions the true line wins 96% of the reports with median peak score 0.93. The short-window variant ranks the true load first in only 2 of 12 load interventions. The learned representation thus does not separate 3–14% load changes from normal operation, whereas the physical signatures carry hundreds to millions of nats per second for the same changes (Fig. 5). The same pattern was already present in their development validation (Section VI), so the comparison shows that the physical bank recovers load interventions that these particular adapters miss; it is not a test of the best attainable learned detector at a common alarm budget.

TABLE 6. **Recorded stage times from the continuous evaluation on 120-s records. Medians and ranges summarize full-record stages; the per-report column is a median-total divided by 3,601 reports and is amortized, not frame latency. Stages were recorded in a campaign with one BLAS and one Julia thread; memory and per-report latency were not recorded. This descriptive table does not establish real-time operation or a speedup under matched implementation conditions.**

Stage	Median (s)	Range (s)	ms/report†
Bank build	80.1	62.0–309.7	22.2
History inference	358.7	218.2–2641.1	99.6
ExtraTrees (both)	9.5	6.7–50.5	2.6
Chen detector	1.3	0.9–10.6	0.4

In the persistent-line test, both banks and both adapters identify the correct line first on all 162 openings. The adapter of Chen *et al.*'s quickest-change detector identifies 74/162; 66 records stop before the opening and 22 afterward on a wrong line, and it stops on all 72 paired normal records (first stop after 2,522 s of cumulative normal exposure, against 42,336 s monitored without alarm by the other methods). Its white-injection, quasi-static model does not describe the colored, dynamically filtered measurements of this plant, so its threshold is far from its nominal false-alarm rate. We therefore read this result as a test of transfer under model mismatch, not as a ranking of quickest-change detection. After the first identification, the two learned adapters raise 164 and 179 further unmatched alarms on the same persistent records, against 0 for the physical bank. Stratified tables, failure ledgers, restoration outcomes and censoring are in the supplement.

B. COMPUTATIONAL COST

Table 6 records the stages of the evaluation campaign, which ran the five-model bank on one thread. Its history inference

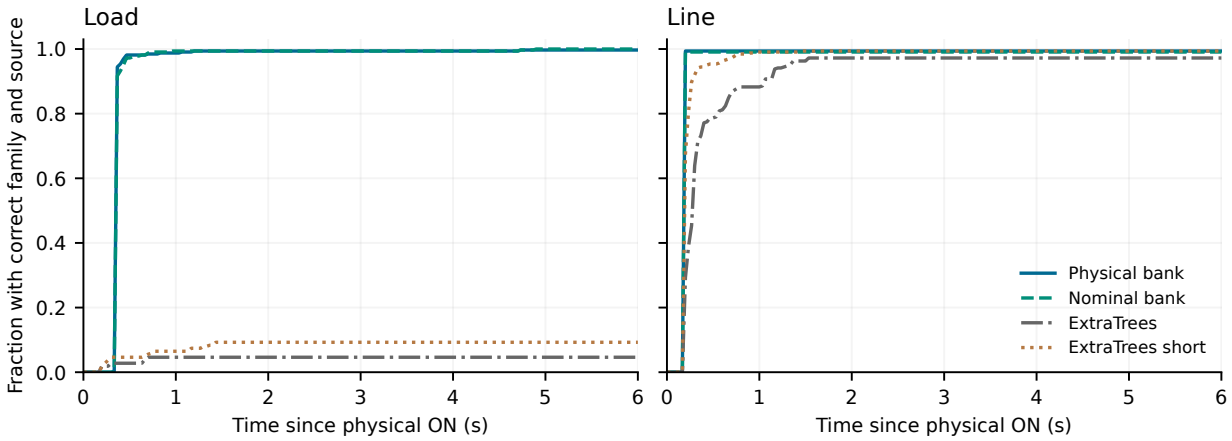


FIGURE 7. **Distribution of correct diagnosis time, using all 324 replays per true family in each denominator. An incorrect or absent diagnosis contributes no mass, and late correct decisions are included, so the curves differ from the before-OFF endpoint.**

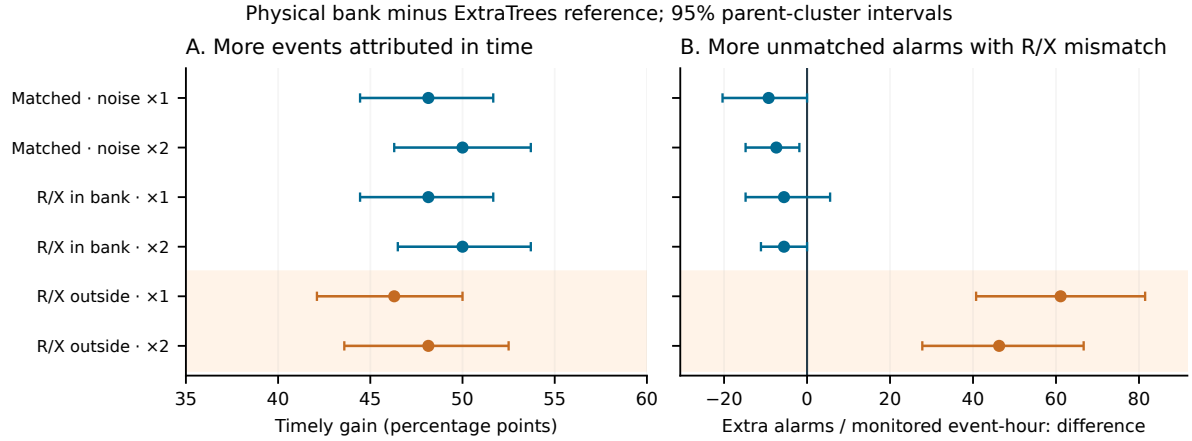


FIGURE 8. **Physical bank minus the ExtraTrees reference in the six paired strata. Points and 95% intervals come from 10,000 physical-parent resamples. Left: percentage-point difference in timely attribution; right: difference in unmatched alarms per monitored event-record hour (the 12-s startup is excluded). Orange rows mark out-of-bank R/X conditions.**

takes a median 358.7 s per 120-s record (99.6 ms per report), three times slower than the 30-report/s input, so the five-model bank as implemented is an offline causal replay. The cost is dominated by the number of network hypotheses. Re-running the unchanged nominal bank alone on the same hardware and thread settings (Section VII-F) gives 26.4 ms per report (median over 6 records, maximum 30.0 ms), below the 33.3-ms report interval; the nominal bank attributed 643/648 interventions in the independent evaluation. These are amortized processing rates on a desktop CPU; per-report latency percentiles and memory were not instrumented, and the 80-s startup bank construction is a one-time cost. Decision timestamps elsewhere in the paper are report times, not wall-clock latency.

C. A CONTINUOUS SEQUENCE AND A DELAYED DECISION

A development trajectory illustrates the timing criterion (Fig. 9). The network keeps moving after OFF, and a later event is evaluated against that predicted response. Under out-of-bank network error, the bus-31 source is identified only after its restoration: correct, but not timely. The active-mass trace, the ΔM decomposition and the development counts are in the supplement.

D. A CONTINUOUS SINGLE-STATE PHYSICAL CONTROL

A development control keeps one committed augmented physical state through restoration instead of competing histories. It uses the same nominal plant and measurement operators, profiles severity over short alternatives of at most 19 reports and applies the same six-report publication clock. On 54 development replays (three parents, three R/X conditions), this single-state generalized likelihood ratio (GLR) detector attains 37/54 timely decisions with 37 extras, against 44/54 and 6 for the nominal history bank and 51/54 and 3 for the five-model bank. The development schedules pack six interventions into 80-s records instead of 120 s, which likely explains why the nominal bank scores lower here than in Table 5. The three procedures also differ in onset search and severity scoring, so this comparison indicates, but does not isolate, the benefit of explicit recovery histories over a single

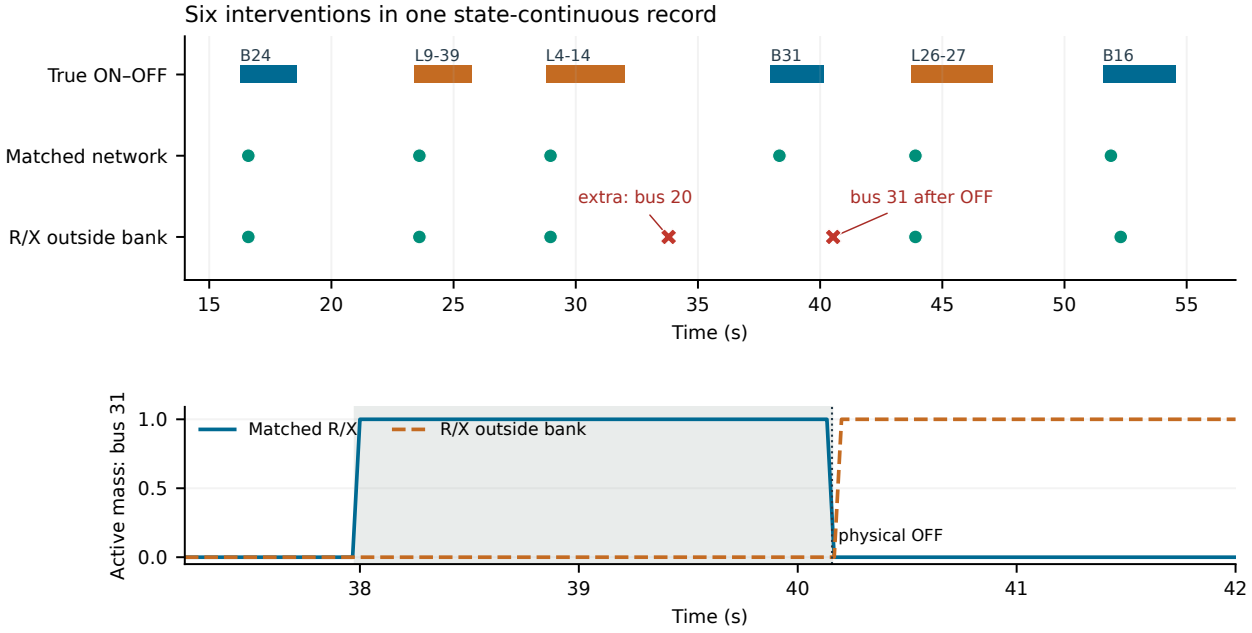


FIGURE 9. **One continuous development trajectory, shown for interpretation. Top: physical ON–OFF intervals and emitted diagnoses under matched and out-of-bank R/X. Bottom: bus-31 active mass with the true intervention shaded. A diagnosis after OFF misses the timely endpoint even when it identifies the source. True timing enters only this retrospective display.**

continuous state.

E. CONTROLLED SEVERITY-SCORE AND STATE-CARRY STUDY

The continuous comparison evaluates complete procedures. To distinguish two mechanisms within the physical bank, we fixed a subsequent 2×2 study: marginal amplitude evidence versus a bounded profile score, and physical-state carry versus removal at inferred closure. The five network models, observations, startup fits, operators, proposal rules and publication thresholds are common to all four arms. Identical thresholds do not imply identical false-alarm rates; both coverage and alarm counts must be measured.

For one load history and amplitude interval $[l, u]$, write its accumulated conditional log likelihood as

$$\ell_k(a) = c_k + b_k a - \tfrac{1}{2} G_k a^2. \tag{25}$$

The original score uses $B_k = \log \int_l^u \exp(\ell_k(a) - c_k) p(a)\, da$ with the existing truncated Gaussian prior. The alternative uses

$$\widehat{a}_k = \mathrm{clip}_{[l,u]}(b_k/G_k), \qquad Q_k = b_k \widehat{a}_k - \tfrac{1}{2} G_k \widehat{a}_k^2, \tag{26}$$

with the constant-likelihood case treated separately. Each report adds $B_k - B_{k-1}$ or $Q_k - Q_{k-1}$, so a rolling replay does not repeatedly add the same suffix evidence. The conditional Gaussian amplitude/state updates remain unchanged. Profile weights are normalized scores, not posterior probabilities. This intervention tests the amplitude-scoring choice, rather than removing every Bayesian calculation. The existing Gaussian sufficient statistics supply $G_k = v_k^{-1} - v_0^{-1}$ and $b_k = \mu_k / v_k$, removing the computational prior precision before profiling. Fixed line-opening amplitudes and the prior weights of amplitude intervals are unchanged.

At an internally inferred closure, the carry arm retains the projected augmented state. The reset arm sets its 114 estimated physical-state deviations to zero and retains instrumental/discrepancy coordinates, covariance regime and network-model weights. This is an estimator ablation: every observed record comes from the same continuous physical trajectory. It does not test removal of all temporal information.

The protocol uses all 18 matched-network continuous parents at the original noise level, plus the first three normal and persistent-line parents by identifier. It was fixed before ablation outcomes, but the underlying records had already been examined. Results are therefore a posthoc mechanism study, with paired intervals over physical parents, not a new independent confirmation.

1) Component Outcomes

Table 7 and Fig. 10 report coverage and alarm cost for the four arms. With the state carried, marginal and bounded-profile severity scores tie on every reported endpoint. Resetting the 114 estimated physical-state deviations at inferred closure reduces timely attribution from 108/108 to 75/108 and raises the unmatched alarms from 1 to 166 under both scores. The reset leaves instrumental state, covariance regime, prior weights, thresholds and publication rules intact, and the simulated plant never resets.

Fig. 11 compares this outcome with Prediction 3. The spurious evidence exceeds $\lambda^\star$ after every intervention, by at least two and up to seven orders of magnitude, and the reset arm is harmed after 84 of the 108 interventions: an extra alarm

TABLE 7. Controlled amplitude-score and physical-state-carry ablation. Same five-model bank and publication rules. All 18 continuous parents (108 events) are retained. Normal controls comprise three 600-s records with 588 s evaluated exposure each (1,764 s total; startup excluded). Delay is conditional on emitted matched diagnoses and must be read with the no-timely-diagnosis count.

Procedure	Timely	Extras	Normal FA	False restorations	No timely	Delay (s)
Marginal + carry	108/108	1	0/1,764 s	1	0/108	0.267
Profile + carry	108/108	1	0/1,764 s	1	0/108	0.267
Marginal + reset	75/108	166	0/1,764 s	130	33/108	0.337
Profile + reset	75/108	166	0/1,764 s	130	33/108	0.337

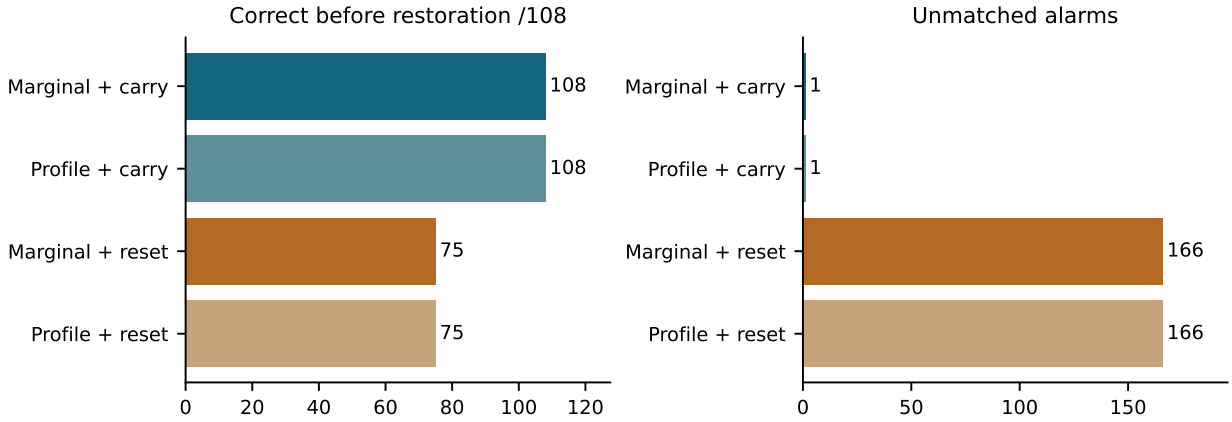


FIGURE 10. Severity-scoring and state-carry study on identical observations and five-model banks. The 108 interventions belong to 18 continuous trajectories. Thresholds are fixed, so the alarm rate may differ between arms.

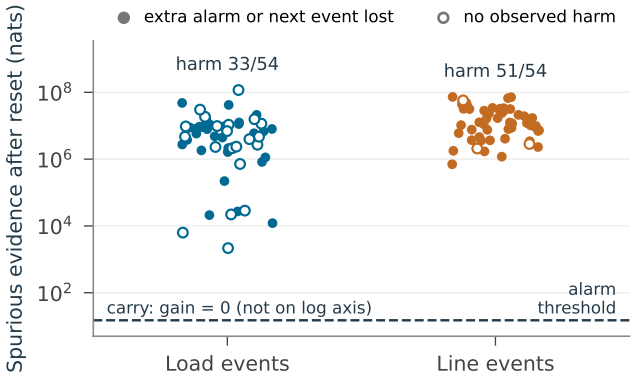


FIGURE 11. Predicted spurious evidence after a state reset, (24), for each of the 108 interventions of the controlled study, against the observed outcome of the reset arm. Filled: an extra alarm in the following gap or a lost timely attribution of the next intervention; open: neither. With carry the predicted spurious evidence is zero.

in the following gap or a lost timely attribution of the next one. Harm is almost certain after line events (51/54), whose openings excite larger electromechanical swings, and occurs after 33/54 load events. The bus-31 events carry the smallest spurious evidence and are harmed after three of seven. The analysis therefore explains why forgetting the state fails at the population level, while the outcome of an individual load event, given evidence far above threshold, is decided by the episode and publication rules rather than by the size of the evidence. These are within-simulator contrasts at the original thresholds; the two scores are compared at equal thresholds, not at equal alarm rates.

F. NOISE STRESS WITH UNCHANGED CODE

To test Predictions 1 and 2 outside the calibrated noise, we reran the nominal bank without any change to its equations, thresholds or rules on four held-out calibration event records (12 load and 12 line interventions, matched network) and two 600-s normal records (0.33 h). Six noise scenarios were generated with the simulator's own noise stream: the original realization scaled by $\times 1$, $\times 5$, $\times 10$ and $\times 20$, and the original plus independent white per-PMU phasor noise with TVE-equivalent standard deviation 0.05% and 0.1%, propagated to frequency and ROCOF by the simulator's 21-report estimator. The $\times 1$ scenario reproduces the original predictions message for message. The bank re-estimates its noise scale and observation discrepancy from the noisy 300-report prefix; the operating point is kept at its nominal-noise estimate because derivative matrices exist only there. The implementation refuses to start when the estimated noise scale leaves its calibrated support of $[0.1, 10]$; for this test only, that bound was widened, and every other line of code is unchanged.

Table 8 and Fig. 12 give the outcomes. Scaling the calibrated noise up to twenty times leaves line attribution intact and loses one load intervention, a 4.1% change at bus 26, at $\times 5$ and $\times 10$ but not at $\times 20$. Prediction 2 classifies that change as detectable at all three levels, so the miss reflects the episode rules rather than missing information. The median load alarm delay moves only from 0.35 to 0.36 s, and no scenario produces an extra alarm on event records or any alarm on the normal records.

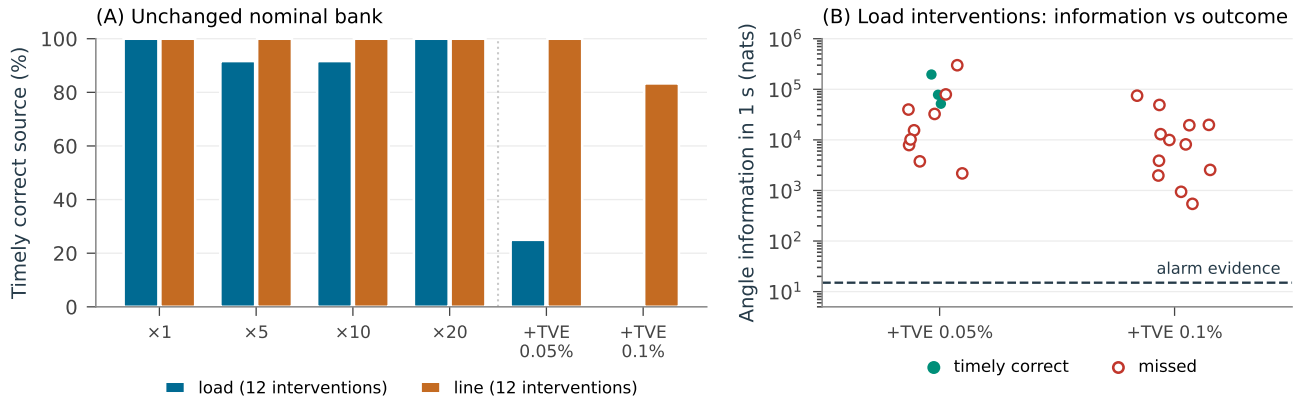


FIGURE 12. **Nominal bank under increased PMU noise, with code, thresholds and rules unchanged. (A) Timely correct family and source by scenario: the calibrated noise scaled by ×1–×20 (left of the dotted line) and the original noise plus independent white per-PMU phasor noise of 0.05% and 0.1% TVE (right). (B) For each load intervention under added independent noise, the information that a matched filter on cross-PMU-centered voltage and current angles accumulates in 1 s, against the observed outcome; the dashed line is the evidence required for an alarm.**

TABLE 8. **Noise-stress outcomes of the unchanged nominal bank: timely correct attributions, unmatched alarms on event records, and alarms on the two normal records.**

Scenario	Load	Line	Extras	Normal
×1	12/12	12/12	0	0
×5	11/12	12/12	0	0
×10	11/12	12/12	0	0
×20	12/12	12/12	0	0
+TVE 0.05%	3/12	12/12	0	0
+TVE 0.1%	0/12	10/12	0	0

Independent per-PMU noise behaves differently: the bank attributes 3/12 load interventions at 0.05% TVE and 0/12 at 0.1%, while lines remain at 12/12 and 10/12. These failures are not information-limited. For the same twelve load interventions, a matched filter on the cross-PMU-centered voltage and current angles, with the added independent noise alone, accumulates at least 2.2×10^3 nats in one second at 0.05% TVE and 5.4×10^2 at 0.1%, 36 times the alarm evidence or more. The implemented filter, however, adapts a single noise scale per signal family and keeps the cross-PMU correlation of the reference data, in which angle noise is almost entirely common to all units. Independent noise therefore raises its estimated angle-noise scale about 48-fold and suppresses the angle information on which load attribution relies; lines, whose signatures are two to three orders of magnitude larger (Fig. 5), survive. The test confirms Predictions 1 and 2 for the noise structure the bank models, and it identifies estimation of the cross-PMU noise covariance from the startup prefix as the extension required for field PMUs.

VIII. DISCUSSION

A. WHAT THE PHYSICS EXPLAINS

The information analysis turns the headline accuracy from an empirical observation into a consequence of the network equations. Under the noise of the reference data, every catalog intervention carries hundreds to tens of millions of times the evidence needed for an alarm within one second, so detection is limited by the publication rule rather than by information. Line alarms arrive at the confirmation floor, as predicted. Load alarms arrive about 4 reports later than the information allows. This latency is not explained by the information and is therefore algorithmic; the likely sources are the onset-proposal screen and the competition among severity intervals, which could be changed without new measurements. The errors that remain occur under out-of-bank R/X, where the predicted signature itself is wrong. The same analysis predicts where attribution should stop working, at the bus-31 load once the noise grows by a factor of about 7.4, and why forgetting the physical state is costly everywhere: the free recovery response projected on the PMUs is two to seven orders of magnitude larger than the alarm evidence. Physical-state continuity is therefore not a tuning choice but a requirement implied by the size of that response.

B. RELATION TO LEARNED AND STATISTICAL DETECTORS

The learned adapters and the physical bank differ in how they obtain source responses: from labeled trajectories, or from the DAE operators. The adapters reproduce the line responses but miss most load changes, a pattern already present in their development validation. The comparison therefore shows that a physically derived dictionary attributes small load changes that these adapters do not, at the cost of more alarms when the network model is wrong. It does not establish superiority over a learned detector calibrated to a common alarm budget. Similarly, the adapter of Chen *et al.* fails here because its quasi-static white-noise model does not match the plant, not because quickest-change detection is unsuitable; an information-matched version of that test would use the switched-DAE signatures of Section V.

C. LIMITS OF THE EVIDENCE

Shared model family. Observations and predictors come from the same research plant. Nonlinear switching, noise and out-

of-bank R/X challenge the linearized inverse model, but transfer to a different dynamic model or to field data is untested. The nine load and nine line candidates were fixed during method development; they form a declared catalog, not a sample of all assets or event families.

Measurement realism. The noise model reproduces the reference data, whose differential angle noise is small. Independent per-PMU noise of 0.05–0.1% TVE, well inside the standard's limits, suppresses most load attributions of the implemented bank even though the angles still carry ample information (Section VII-F). Field use therefore requires estimating the cross-PMU noise covariance, not only its scale, from the startup prefix; this extension was not tested here. The terminal map retains a provisional BUS5 current terminal and excludes the BUS39 current angle; it is a research measurement hypothesis, not a verified installation. Cross-family noise covariance is not applied.

Approximation. Pruning, steady-state covariances, plug-in startup estimates and mean compression at closure make the bank approximate; its masses are attribution weights, not calibrated probabilities. Retrospective proposals update the history posterior but not the network-model weights. The birth and restoration hazards are conservative relative to the test schedules; their effect is bounded by (22) but was not swept.

Operation. The startup prefix must be event-free, all PMUs available and the noise within the implemented support. The nominal bank runs faster than the input on one thread; the five-model bank does not. These results support diagnostic development, not protection or control deployment.

IX. CONCLUSION

A restored electrical intervention can remain the cause of the observed response. We formulated load/line attribution from sparse PMUs as change detection and isolation with switched-DAE post-change models, in which histories start from estimated past states, keep that state through restoration and separate the active intervention from its originating cause. Because the histories are linear-Gaussian given their mode paths, the network operators also give each candidate's detection and isolation information, its minimum detectable change and the cost of forgetting the physical state.

In an independent simulated evaluation the bank attributes 644/648 interventions before restoration and raises no alarm in 11.8 h of normal records; its remaining errors and most of its 72 extra alarms arise from network-model mismatch. The information analysis explains this level of performance, separates the information-limited part of the alarm delay from its algorithmic part, and predicts the cost of resetting the state and the noise level at which the smallest load becomes undetectable. With unchanged code the bank tolerates a twentyfold increase of the calibrated noise. Independent per-PMU noise defeats its scalar noise adaptation while the angle signatures still carry ample information, so the cross-PMU noise covariance, not the physics, is the next element to estimate. The single-model bank already runs faster than the PMU reporting rate.

Two directions follow. The detectability analysis can be extended from a fixed catalog to placement and catalog design, since it ranks sources by information before any data are collected. The same framework accommodates other parameter-override events, such as generation changes, and measurement-integrity hypotheses that are inconsistent with network dynamics; testing them, together with transfer to an independent dynamic model, is the next step toward field use.

REFERENCES

[1] I. V. Nikiforov, "A generalized change detection problem," *IEEE Transactions on Information Theory*, vol. 41, no. 1, pp. 171–187, 1995.

[2] T. L. Lai, "Sequential multiple hypothesis testing and efficient fault detection-isolation in stochastic systems," *IEEE Transactions on Information Theory*, vol. 46, no. 2, pp. 595–608, 2000.

[3] A. Tartakovsky, I. Nikiforov, and M. Basseville, *Sequential Analysis: Hypothesis Testing and Changepoint Detection*. Boca Raton, FL: CRC Press, 2014.

[4] A. S. Willsky and H. L. Jones, "A generalized likelihood ratio approach to the detection and estimation of jumps in linear systems," *IEEE Transactions on Automatic Control*, vol. 21, no. 1, pp. 108–112, 1976.

[5] J. E. Tate and T. J. Overbye, "Line outage detection using phasor angle measurements," *IEEE Transactions on Power Systems*, vol. 23, no. 4, pp. 1644–1652, 2008.

[6] Y. C. Chen, T. Banerjee, A. D. Domínguez-García, and V. V. Veeravalli, "Quickest line outage detection and identification," *IEEE Transactions on Power Systems*, vol. 31, no. 1, pp. 749–758, 2016.

[7] G. Rovatsos, X. Jiang, A. D. Domínguez-García, and V. V. Veeravalli, "Statistical power system line outage detection under transient dynamics," *IEEE Transactions on Signal Processing*, vol. 65, no. 11, pp. 2787–2797, 2017.

[8] A. Warner, D. Hamilton, G. Fellouris, and D. Aliprantis, "Quick line fault detection and identification using high-fidelity models of transient dynamics," TechRxiv preprint, version 1, 2023.

[9] A. N. Shiryaev, "On optimum methods in quickest detection problems," *Theory of Probability and Its Applications*, vol. 8, no. 1, pp. 22–46, 1963.

[10] B. Friedland, "Treatment of bias in recursive filtering," *IEEE Transactions on Automatic Control*, vol. 14, no. 4, pp. 359–367, 1969.

[11] H. A. P. Blom and Y. Bar-Shalom, "The interacting multiple model algorithm for systems with markovian switching coefficients," *IEEE Transactions on Automatic Control*, vol. 33, no. 8, pp. 780–783, 1988.

[12] Y. Boers and H. Driessen, "A multiple model multiple hypothesis filter for Markovian switching systems," *Automatica*, vol. 41, no. 4, pp. 709–716, 2005.

[13] S. Gillijns and B. D. Moor, "Unbiased minimum-variance input and state estimation for linear discrete-time systems," *Automatica*, vol. 43, no. 1, pp. 111–116, 2007.

[14] S. A. Nugroho, A. F. Taha, N. Gatsis, and J. Zhao, "Observers for differential algebraic equation models of power networks: Jointly estimating dynamic and algebraic states," *IEEE Transactions on Control of Network Systems*, vol. 9, no. 3, pp. 1531–1543, 2022.

[15] M. Katanic, J. Lygeros, and G. Hug, "Recursive dynamic state estimation for power systems with an incomplete nonlinear DAE model," *IET Generation, Transmission & Distribution*, vol. 18, no. 22, pp. 3657–3668, 2024.

[16] T. L. Fernando, H. M. Trinh, and L. Jennings, "Functional observability and the design of minimum order linear functional observers," *IEEE Transactions on Automatic Control*, vol. 55, no. 5, pp. 1268–1273, 2010.

[17] A. N. Montanari, L. Freitas, D. Proverbio, and J. Gonçalves, "Functional observability and subspace reconstruction in nonlinear systems," *Physical Review Research*, vol. 4, no. 4, p. 043195, 2022.

[18] D. Eriksson, E. Frisk, and M. Krysander, "A method for quantitative fault diagnosability analysis of stochastic linear descriptor models," *Automatica*, vol. 49, no. 6, pp. 1591–1600, 2013.

[19] L. L. Scharf and B. Friedlander, "Matched subspace detectors," *IEEE Transactions on Signal Processing*, vol. 42, no. 8, pp. 2146–2157, 1994.

[20] Y. Zhao, J. Chen, A. Goldsmith, and H. V. Poor, "Identification of outages in power systems with uncertain states and optimal sensor locations," *IEEE*

Journal of Selected Topics in Signal Processing, vol. 8, no. 6, pp. 1140–1153, 2014.
[21] R. Anguluri, O. Kosut, and L. Sankar, "Localization and estimation of unknown forced inputs: A group LASSO approach," *IEEE Transactions on Control of Network Systems*, vol. 10, no. 4, pp. 1997–2009, 2023.
[22] M. Jamei, R. Ramakrishna, T. Tesfay, R. Gentz, C. Roberts, A. Scaglione, and S. Peisert, "Phasor measurement units optimal placement and performance limits for fault localization," *IEEE Journal on Selected Areas in Communications*, vol. 38, no. 1, pp. 180–192, 2020.
[23] F. Conte, F. D'Agostino, B. Gabriele, G.-P. Schiapparelli, and F. Silvestro, "Fault detection and localization in active distribution networks using optimally placed phasor measurements units," *IEEE Transactions on Power Systems*, vol. 38, no. 1, pp. 714–727, 2023.
[24] T. Yildiz and A. Abur, "Sparse PMU placement algorithm for enhanced detection and identification of power grid events," *IEEE Transactions on Power Systems*, vol. 40, no. 2, pp. 1843–1853, 2025.
[25] A. Warner and G. Fellouris, "Sequential change diagnosis revisited and the adaptive matrix CuSum," *Bernoulli*, vol. 30, no. 3, pp. 2228–2252, 2024.
[26] R. Nagi, X. Huan, and Y. C. Chen, "Bayesian inference of parameters in power system dynamic models using trajectory sensitivities," *IEEE Transactions on Power Systems*, vol. 37, no. 2, pp. 1253–1263, 2022.
[27] A. Alalem Albustami, A. F. Taha, and S. Mahadevan, "DAE-aware bayesian inference for joint generator-network parameter estimation," In press, Electric Power Systems Research; arXiv:2604.15686, 2026. [Online]. Available: https://arxiv.org/abs/2604.15686
[28] S. Pandey, A. K. Srivastava, and B. G. Amidan, "A real time event detection, classification and localization using synchrophasor data," *IEEE Transactions on Power Systems*, vol. 35, no. 6, pp. 4421–4431, 2020.
[29] Z. Li, J. Tian, and H. Liu, "A joint data and knowledge-driven method for power system disturbance localisation," *IET Generation, Transmission & Distribution*, vol. 18, no. 24, pp. 4078–4089, 2024.
[30] T. Yildiz and A. Abur, "Convolutional neural network-assisted fault detection and location using few PMUs," *Electric Power Systems Research*, vol. 235, p. 110705, 2024.
[31] A. Pomarico, A. Berizzi, and J. N. Kutz, "A shallow recurrent decoder for dynamic state estimation with a limited number of PMUs in power systems," *arXiv preprint arXiv:2607.00116*, 2026.
[32] A. Dey, S. Chakraborty, and M. V. Salapaka, "Topology change estimation in power distribution networks: A machine learning approach," *IEEE Transactions on Smart Grid*, vol. 17, no. 4, 2026.
[33] IEEE, "IEEE standard for synchrophasor measurements for power systems," IEEE Std C37.118.1-2011, 2011.



MASOUD BARATI received the Ph.D. degree in electrical engineering from the Illinois Institute of Technology in 2013. He is with the Department of Electrical and Computer Engineering, University of Pittsburgh, Pittsburgh, PA, USA. His research includes mathematical modeling and optimization of electric power systems.



PAYMAN DEHGHANIAN received the Ph.D. degree from Texas A&M University in 2017. He is an Associate Professor with the Department of Electrical and Computer Engineering, The George Washington University, Washington, DC, USA. His research includes power-system reliability, resilience, and data-informed decision-making.

JORGE LUIS MAYORGA TABORDA studied electronic engineering at Universidad de los Andes, Bogotá, Colombia. His academic work has addressed distributed control using population dynamics, including applications to urban traffic systems.



DAVID FELIPE CELEITA RODRÍGUEZ received the Ph.D. degree in engineering from Universidad de los Andes, Bogotá, Colombia. He is an Associate Professor and Director of Outreach and Engagement with the Faculty of Engineering, Universidad de La Sabana, Chía, Colombia. His research includes power-system protection, fault location, and real-time simulation.

YESSICA AFRICANO RODRIGUEZ carried out undergraduate thesis work on photovoltaic integration at Universidad de los Andes, Bogotá, Colombia. Her published research includes co-simulation and stochastic assessment of photovoltaic hosting capacity in distribution systems.